recursion can always be easily written as a simple loop. For example:

Listing E.7 Factorial function, iterative version

```
function factorial(n) {
  let res = 1;                                          #1
  if (n < 0) {
    throw new RangeError('n can not be negative');
  }

  while (n > 1)
{                                                       #2
    res *= n--;
  }
  return res;
}
```

#1 Base case
#2 Simulate tail recursion with a loop

Mutual tail recursion can also be optimized in the same way, but most compiler will only optimize simple tail recursion.

E.1.4 Mutual Recursion

A function can call itself directly, but also through another function. If 2 or more functions call themselves in a cycle, then they are *mutual recursive*.

Listing E.8 An example of mutual recursive functions

```
function f(n) {
```